1.19 A closed vessel having a volume of 1 liter holds 1.2×10^{22} molecules of nitrogen gas. For the nitrogen, determine (a) the amounts present, in kmol and kg, and (b) the specific volumes, in m^3/kmol and m^3/kg.

KNOWN: A vessel of known volume holds a specified number of molecules of nitrogen gas.

FIND: Determine (a) the mass and number of moles present, and (b) the specific volumes on molar and mass bases.

SCHEMATIC AND GIVEN DATA:

$g = 9.81$ m/s^2
$M_{\text{nitrogen}} = 28.01$ kg/kmol (Table A-1)

1.2×10^{22} molecules
nitrogen gas
$V = 1$ liter

ENGINEERING MODEL:
1. Nitrogen is a closed system.

ANALYSIS:
(a) From Section 1.5, the number of molecules in a gram mole (mol) is 6.022×10^{23} (Avogadro's number). Thus

$$n = \frac{1.2\times10^{22}\text{ molecules}}{6.022\times10^{23}\,\dfrac{\text{molecules}}{\text{mol}}} = 0.0199\text{ mol}$$

Converting to kmol

$$n = (0.0199\text{ mol})\left|\frac{1\text{ kmol}}{10^3\text{ mol}}\right| = \underline{\mathbf{1.99\times10^{-5}\ kmol}}$$

Using Eq. 1.8 to determine the mass of the nitrogen

$$m = nM = (1.99\times10^{-5}\text{ kmol})\left(28.01\frac{\text{kg}}{\text{kmol}}\right) = \underline{\mathbf{5.57\times10^{-4}\ kg}}$$

(b) Specific volume on a molar basis is

$$\bar{v} = \frac{V}{n} = \frac{1\text{ liter}}{1.99\times10^{-5}\text{ kmol}}\left|\frac{10^{-3}\text{ m}^3}{1\text{ liter}}\right| = \underline{\mathbf{50.25\ m^3/kmol}}$$

Specific volume on a mass basis is

$$v = \frac{V}{m} = \frac{1 \text{ liter}}{5.57 \times 10^{-4} \text{ kg}} \left| \frac{10^{-3} \text{ m}}{1 \text{ liter}} \right| = \underline{\mathbf{1.80 \ m^3/kg}}$$

1.20 The specific volume of 5 kg of water vapor at 1.5 MPa, 440°C is 0.2160 m^3/kg. Determine (a) the volume, in m^3, occupied by the water vapor, (b) the amount of water vapor present, in gram moles, and (c) the number of molecules.

KNOWN: Mass, pressure, temperature, and specific volume of water vapor.

FIND: Determine (a) the volume, in m^3, occupied by the water vapor, (b) the amount of water vapor present, in gram moles, and (c) the number of molecules.

SCHEMATIC AND GIVEN DATA:

$m = 5$ kg
$p = 1.5$ MPa
$T = 440$°C
$v = 0.2160$ m^3/kg

ENGINEERING MODEL:

1. The water vapor is a closed system.

ANALYSIS:

(a) The specific volume is volume per unit mass. Thus, the volume occupied by the water vapor can be determined by multiplying its mass by its specific volume.

$$V = mv = (5\text{ kg})\left(0.2160\frac{\text{m}^3}{\text{kg}}\right) = \underline{\mathbf{1.08\ m^3}}$$

(b) Using molecular weight of water from Table A-1 and applying the appropriate relation to convert the water vapor mass to gram moles gives

$$n = \frac{m}{M} = \left(\frac{5\text{ kg}}{18.02\frac{\text{kg}}{\text{kmol}}}\right)\left|\frac{1000\text{ moles}}{1\text{ kmol}}\right| = \underline{\mathbf{277.5\ moles}}$$

(c) Using Avogadro's number to determine the number of molecules yields

$$\#\,\text{Molecules} = \text{Avogadro's Number} \times \#\,\text{moles} = \left(6.022\times10^{23}\,\frac{\text{molecules}}{\text{mole}}\right)(277.5\text{ moles})$$

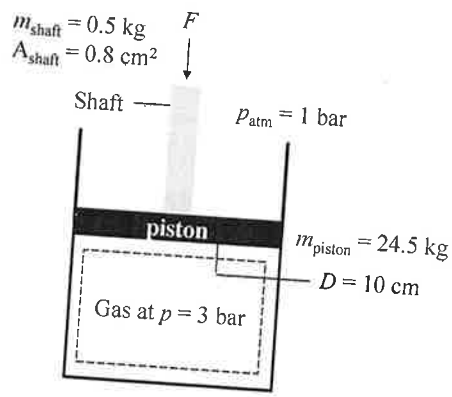
$$\#\,\text{Molecules} = \underline{\mathbf{1.671\times10^{26}\ molecules}}$$

1.24 Figure P1.24 shows a gas contained in a vertical piston-cylinder assembly. A vertical shaft whose cross-sectional area is 0.8 cm^2 is attached to the top of the piston. Determine the magnitude, F, of the force acting on the shaft, in N, required if the gas pressure is 3 bar. The masses of the piston and attached shaft are 24.5 kg and 0.5 kg, respectively. The piston diameter is 10 cm. The local atmospheric pressure is 1 bar. The piston moves smoothly in the cylinder and $g = 9.81$ m/s^2.

KNOWN: A piston-cylinder assembly with a vertical shaft attached to the piston contains gas.

FIND: The required magnitude of the force acting on the shaft if the gas is at a specified pressure.

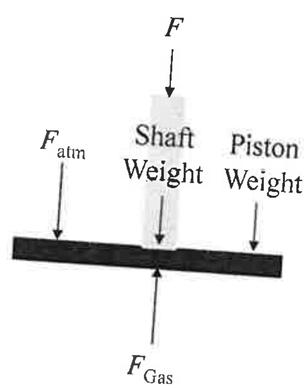
SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL:
1. The gas is a closed system.
2. The piston is in static equilibrium.
3. Atmospheric pressure is exerted on the top of the piston.
4. Local gravitational acceleration is 9.81 m/s^2.

ANALYSIS:

Since the piston moves smoothly within the cylinder, the force exerted by the gas equals the resisting force comprised of the piston weight, shaft weight, the force exerted by the atmospheric pressure, and the force acting on the shaft, F. That is, the sum of the forces acting vertically is zero, giving



$$F_{gas} = \text{Piston Weight} + \text{Shaft Weight} + F_{atm} + F$$

Solving,

$$F = F_{gas} - \text{Piston Weight} - \text{Shaft Weight} - F_{atm} \quad (*)$$

In this expression,

$F_{gas} = p_{gas}A_{piston}$, where A_{piston} is the piston force area:

$$A_{piston} = \frac{\pi D^2_{piston}}{4} = \frac{\pi(10\text{ cm})^2}{4} = 78.54\text{ cm}^2$$

Therefore, $F_{gas} = (3\text{ bar})\left|\frac{10^5 \frac{\text{N}}{\text{m}^2}}{1\text{ bar}}\right|(78.54\text{ cm}^2)\left|\frac{1\text{ m}}{10^2\text{ cm}}\right|^2 = 2356.2\text{ N}$

The pressure of the atmosphere acts only on the net area at the top of the piston – namely, the piston face area less the area occupied by the shaft. The force is then

$F_{atm} = p_{atm}(A_{piston} - A_{shaft})$

$$F_{atm} = (1\text{ bar})(78.54\text{ cm}^2 - 0.8\text{ cm}^2)\left|\frac{10^5 \frac{\text{N}}{\text{m}^2}}{1\text{ bar}}\right|\left|\frac{1\text{ m}}{10^2\text{ cm}}\right|^2 = 777.4\text{ N}$$

The total weight of the piston and shaft is

Total Weight = $(m_{piston} + m_{shaft})g$

$$\text{Total Weight} = (24.5\text{ kg} + 0.5\text{ kg})\left(9.81\frac{\text{m}}{\text{s}^2}\right)\left|\frac{1\text{ N}}{1\frac{\text{kg}\cdot\text{m}}{\text{s}^2}}\right| = 245.3\text{ N}$$

Collecting results, Eq. (*) gives

$$F = 2356.2\text{ N} - 245.3\text{ N} - 777.4\text{ N} = \underline{\mathbf{1333.5\text{ N}}}$$

1.25 A gas contained within a piston-cylinder assembly undergoes four processes in series:

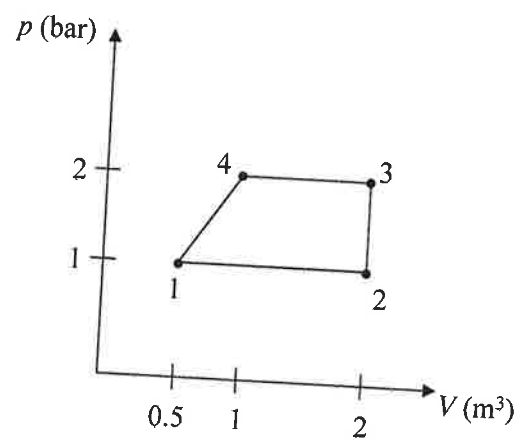
Process 1-2: Constant-pressure expansion at 1 bar from $V_1 = 0.5\ \mathrm{m}^3$ to $V_2 = 2\ \mathrm{m}^3$

Process 2-3: Constant volume to 2 bar

Process 3-4: Constant-pressure compression to 1 m^3

Process 4-1: Compression with $pV^{-1} = constant$

Sketch the process in series on a *p-V* diagram labeled with pressure and volume values at each numbered state.



1.26 Referring to Fig. 1.7,
(a) if the pressure in the tank is 1.5 bar and atmospheric pressure is 1 bar, determine L, in m, for water with a density of 997 kg/m^3 as the manometer liquid. Let $g = 9.81$ m/s^2.
(b) determine L, in cm, if the manometer liquid is mercury with a density of 13.59 g/cm^3 and the gas pressure is 1.3 bar. A barometer indicates the local atmospheric pressure is 750 mmHg. Let $g = 9.81$ m/s^2.

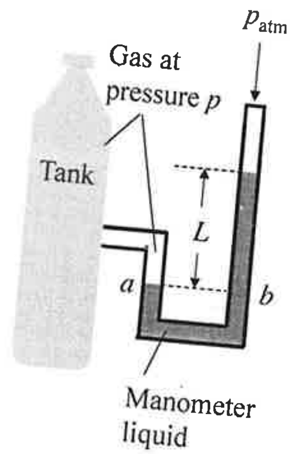
KNOWN: A manometer is attached to a tank containing a gas.

FIND: L considering two different manometer liquids with associated gas pressures.

SCHEMATIC AND GIVEN DATA:

(a) $p_{gas} = 1.5$ bar
$p_{atm} = 1$ bar
$\rho_{water} = 997$ kg/m^3

(b) $p_{gas} = 1.3$ bar
$p_{atm} = 750$ mmHg
$\rho_{mercury} = 13.59$ g/cm^3



ENGINEERING MODEL:
1. Local gravitational acceleration is 9.81 m/s^2.

ANALYSIS:
(a) We have $p_a = p_{gas}$ and $p_a = p_b$. p_b is evaluated using Eq. 1.11. Collecting results,

$$p_{gas} = p_{atm} + \rho_w g L$$

where $\rho_w = 997$ kg/m^3 and $g = 9.81$ m/s^2.

Solving for L

$$L = \frac{p_{gas} - p_{atm}}{\rho_w g} = \frac{(1.5-1)\text{ bar}}{\left(997\dfrac{\text{kg}}{\text{m}^3}\right)\left(9.81\dfrac{\text{m}}{\text{s}^2}\right)}\left|\frac{10^5\dfrac{\text{N}}{\text{m}^2}}{1\text{ bar}}\right|\left|\frac{1\dfrac{\text{kg}\cdot\text{m}}{\text{s}^2}}{1\text{ N}}\right| = \underline{\mathbf{5.11\text{ m}}}$$

(b) First solve for p_{atm} with $L = 750$ mmHg and $\rho_{mercury} = 13.59$ g/cm^3.

$$p_{atm} = \rho_{mercury} g L$$

$$p_{atm} = \left(13.59\frac{\text{g}}{\text{cm}^3}\right)\left|\frac{1\text{ kg}}{10^3\text{ g}}\right|\left|\frac{10^2\text{ cm}}{1\text{ m}}\right|^3\left(9.81\frac{\text{m}}{\text{s}^2}\right)(750\text{ mmHg})\left|\frac{1\text{ m}}{10^3\text{ mm}}\right|\left|\frac{1\text{ N}}{1\frac{\text{kg}\cdot\text{m}}{\text{s}^2}}\right| = 10^5\text{ N/m}^2$$

Following the approach of part (a) and solving for L

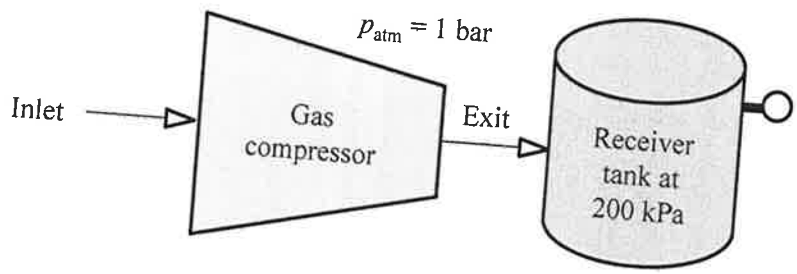
$$L = \frac{p_{gas} - p_{atm}}{\rho_w g} = \frac{(1.3-1)\text{ bar}}{\left(13.59\frac{\text{g}}{\text{cm}^3}\right)\left(9.81\frac{\text{m}}{\text{s}^2}\right)}\left|\frac{10^3\text{g}}{1\text{ kg}}\right|\left|\frac{1\text{ m}}{10^2\text{cm}}\right|^3\left|\frac{10^5\frac{\text{N}}{\text{m}^2}}{1\text{ bar}}\right|\left|\frac{1\frac{\text{kg}\cdot\text{m}}{\text{s}^2}}{1\text{ N}}\right|\left|\frac{100\text{ cm}}{1\text{ m}}\right| = \mathbf{\underline{22.5\text{ cm}}}$$

1.28 As shown in Figure P1.28, the exit of a gas compressor empties into a receiver tank, maintaining the tank contents at a pressure of 200 kPa. If the local atmospheric pressure is 1 bar, what is the reading of the Bourdon gage mounted on the tank wall in kPa? Is this a *vacuum* pressure or a *gage* pressure? Explain.

KNOWN: The exit of a gas compressor empties into a receiver tank.

FIND: The Bourdon gage reading. Identify whether the reading is a *vacuum* pressure or a *gage* pressure and explain.

SCHEMATIC AND GIVEN DATA:



ANALYSIS:
Converting the local atmospheric pressure to kPa, we get p_{atm} = 100 kPa. Since the pressure in the tank, 200 kPa, is greater than the atmospheric pressure, **the Bourdon reading is a *gage* pressure**. Using the relationship

$$p_{gage} = p_{abs} - p_{atm} = 200 \text{ kPa} - 100 \text{ kPa} = 100 \text{ kPa}$$

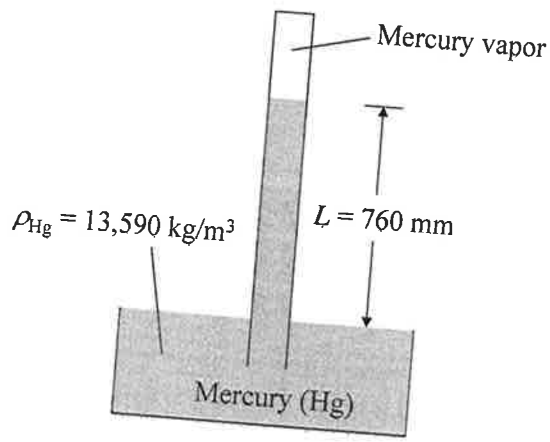
The Bourdon reading is 100 kPa.

1.32 Show that a standard atmospheric pressure of 760 mmHg is equivalent to 101.3 kPa. The density of mercury is 13,590 kg/m^3 and $g = 9.81$ m/s^2.

KNOWN: Standard atmospheric pressure of 760 mmHg.

FIND: Show that 760 mmHg is equivalent to 101.3 kPa.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL:

1. Local gravitational acceleration is 9.81 m/s^2.
2. Pressure of mercury vapor is much less than that of the atmosphere and can be neglected.

ANALYSIS:

Equation 1.12 applies.

$$p_{atm} = p_{vapor} + \rho_{Hg} g L = \rho_{Hg} g L$$

Neglecting the pressure of mercury vapor and applying appropriate conversion factors yield

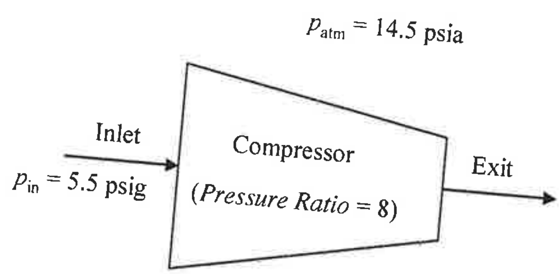
$$p_{atm} = \left(13{,}590\frac{kg}{m^3}\right)\left(9.81\frac{m}{s^2}\right)(760\text{ mm})\left|\frac{1\text{ N}}{1\frac{kg\cdot m}{s^2}}\right|\left|\frac{1\text{ m}}{1000\text{ mm}}\right|\left|\frac{1\text{ kPa}}{1000\frac{N}{m^2}}\right| = \underline{\mathbf{101.3\text{ kPa}}}$$

1.33 A gas enters a compressor that provides a pressure ratio (exit pressure to inlet pressure) equal to 8. If a gage indicates the gas pressure at the inlet is 5.5 psig, what is the absolute pressure, in psia, of the gas at the exit? Atmospheric pressure is 14.5 lbf/in.^{2}

KNOWN: Gas pressure is measured at the inlet of a compressor for which the pressure ratio is known.

FIND: Determine the absolute pressure of the gas at the compressor exit.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL:

1. Atmospheric pressure is 14.5 lbf/in.^{2}

ANALYSIS:

From the compressor pressure ratio, the exit pressure can be determined from

$$\text{pressure ratio} = p_{exit}/p_{in} \quad \rightarrow \quad p_{exit} = p_{in}(\text{pressure ratio})$$

Inlet pressure must be expressed as absolute pressure to solve for exit pressure. Conversion from the inlet pressure gage reading to absolute pressure is determined from

$$p_{in}(\text{gage}) = p_{in}(\text{absolute}) - p_{atm}(\text{absolute})$$

Rearranging the equation to solve for p_{in}(absolute) and substituting values yield

$$p_{in}(\text{absolute}) = p_{in}(\text{gage}) + p_{atm}(\text{absolute}) = 5.5\text{ psig} + 14.5\text{ psia} = 20\text{ psia}$$

Substituting absolute pressure at the inlet into the equation for exit pressure yields

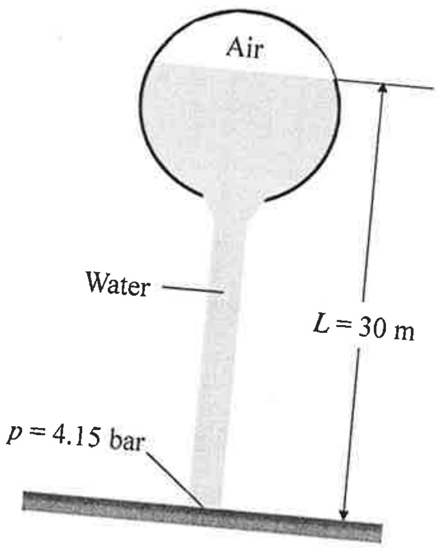
$$p_{exit} = (20\text{ psia})(8) = \underline{\mathbf{160\text{ psia}}}$$

1.37 If the water pressure at the base of the water tower shown in Fig. P1.37 is 4.15 bar, determine the pressure of the air trapped above the water level, in bar. The density of the water is 10^3 kg/m^3. And $g = 9.81$ m/s^2.

KNOWN: Air is trapped above a column of water in a water tower.

FIND: the pressure of the air trapped above the water level.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL:
1. Water density is 10^3 kg/m^3.
2. Local gravitational acceleration is 9.81 m/s^2.

ANALYSIS: Ignoring the vertical variation in pressure of the air trapped above the water level,

$$\underbrace{p}_{\text{pressure at the base}} = p_{\text{air}} + \rho g L$$

$$p_{\text{air}} = p - \rho g L$$

$$p_{\text{air}} = (4.15 \text{ bar}) - \left(10^3 \frac{\text{kg}}{\text{m}^3}\right)\left(9.81 \frac{\text{m}}{\text{s}^2}\right)(30 \text{ m}) \left| \frac{1 \text{ N}}{1 \frac{\text{kg}\cdot\text{m}}{\text{s}^2}} \right| \left| \frac{1 \text{ bar}}{10^5 \frac{\text{N}}{\text{m}^2}} \right| = \underline{\mathbf{1.21 \text{ bar}}}$$